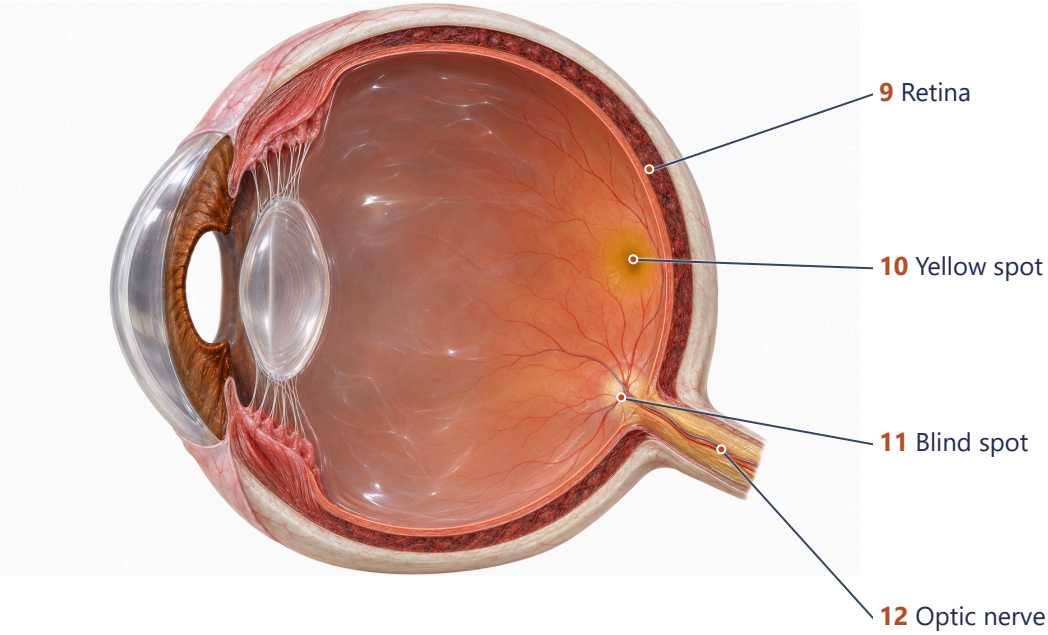
Eye: detecting light

Stimuli and Responses in Humans



Retina + optic nerve

The retina produces nerve impulses. The _____ carries them to the brain.

Yellow spot + blind spot

The yellow spot gives sharp vision. The blind spot has no photoreceptors.

Rods

Rod cells help us see in _____ light. They do not distinguish colours.

Cones

Cone cells support colour vision in bright light. Three types respond best to different ranges of colour.

Quick check

1. Why is a coloured shirt harder to identify in dim light?

A. Rods detect dim light but not colour.

B. The iris produces colours.

C. The blind spot becomes the whole retina.

D. Cones become more sensitive than rods in dim light.

2. Which retinal cells detect colour?

A. Rod cells

B. Cone cells

C. Nerve fibres

D. Iris muscles

3. Why does light falling on the blind spot produce no image signal?

A. The lens is opaque.

B. The pupil closes.

C. It has no photoreceptors.

D. The cornea absorbs all light.

4. Which retinal region gives the sharpest vision?

A. Yellow spot

B. Blind spot

C. Optic nerve

D. Choroid